

Arithmetic Spectral Jets

Stieltjes residues, Gamma reciprocity,
and polylogarithmic untwisting

Research continuation prepared for Vladimir Reshetnikov

Mathematical derivations and computational checks assisted by ChatGPT

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Abstract

The all-order local calculus for polynomial spectral multiplicities extends to arithmetic weights, but ordinary derivatives at the origin generally cease to exist. We prove a finite-pole transfer theorem that expresses every normalized Laurent jet as a polynomial in the factor exponents. Its terms beyond the ordinary Taylor degree are determined entirely by the principal parts of the base Dirichlet series. This answers the explicitly posed arithmetic-multiplicity extension for quasi-polynomial weights and for products of Hurwitz zeta functions, including divisor multiplicities and several distinct poles.

For multiplicative Hurwitz lattices, all local coefficients are finite polynomials in generalized Stieltjes constants. Alternation gives exact algebraic identities between Laurent jets: a six-term first-jet identity for double-divisor multiplicities equals a Vandermonde divided by nine. A convergent centered Gamma series is symmetric in two independent Hurwitz parameters; its derivatives and anchored primitive yield reciprocal polygamma and negative-zeta-derivative identities. Generalized harmonic numbers describe every higher single-shift jet. Polylogarithmic twists lower the pole order, and a finite, explicit Gamma–Stieltjes compensation restores commutation of Abel untwisting with Laurent coefficient extraction, at every order and for every polynomial multiplicity.

All theorem statements have ordinary analytic or algebraic proofs. Exact finite regression tests and independent high-precision diagnostics are supplied separately. The first multiplicative anomaly and the classical special-function ingredients retain their attribution. No claim of worldwide priority, arithmetic independence, or proof-assistant formalization is made.

Source anchor. Targeted repository inspection is pinned to
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Scope. The report addresses question Q10 in the existing *Resonant Gamma Jets and Directional Zeta Identities* continuation. It does not settle the Gaussian S_6 conjecture or the distinct current S_8 conjecture.

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1 Research target, contribution, and conventions

1.1 A specific repository question

The manuscript’s Hurwitz-jet chapters organize parameter derivatives, spectral derivatives, generalized Stieltjes constants, and negative-order polygamma in one calculus [1]. A later report already proves a normalized-exponent polynomial law for polynomial spectral multiplicities, including a polynomial times a power of a logarithm [2]. Its question Q10 asks for arithmetic and nonpolynomial multiplicities, beginning with periodic or quasi-polynomial weights and then moving to base zeta functions with several or higher-order poles.

That is the precise target here. The extension is not obtained by assuming that an arbitrary arithmetic Dirichlet series continues meromorphically. Instead, we state such a continuation as a hypothesis in the general theorem, prove the requisite normal convergence, and verify the hypotheses unconditionally in explicit families. For these families the result is a complete coefficient algorithm rather than an asymptotic extrapolation.

The new scope relative to [2] consists of a general finite-principal-part transfer theorem, the separation of the mean and oscillatory parts of quasi-polynomial multiplicities, independent Hurwitz factors with all their Stieltjes residue layers, explicit Gamma reciprocity and primitive formulas, and an all-order finite compensation for removing a polylogarithmic twist. The alternating identities extend the report’s simple-pole alternants to actual arithmetic higher poles. The argument is elementary once the correct germ is retained; its simplicity is not a reason to identify a Laurent coefficient with an undefined derivative.

1.2 Classical inputs and the limits of a novelty assertion

The Hurwitz continuation, its parameter derivative, and its relation to $\log \Gamma$ are standard [3]. The Pochhammer expansion, the Gamma product, and the polylogarithm expansion near unity are also classical [4, 5]. First-derivative multiplicative-anomaly formulas have an extensive prior theory; Dowker’s treatment includes products of linear factors and polynomial degeneracies [7]. We use that theory as context, not as a new result. The present proofs establish the stated extensions relative to the inspected repository. The literature inspection does not prove that every individual consequence is absent from all earlier publications.

The source audit is targeted. The canonical chapters, the relevant earlier spectral-product section and its research questions, and the incoming-directory inventory were inspected. The incoming ZIP files were not all extracted and audited. The artifact records the source revision and the precise question being answered; it does not assert a line-by-line audit of the entire repository.

1.3 Laurent jets and local coefficients

Throughout, $[s^r]F(s)$ denotes a Laurent coefficient. When $r \geq 0$, define

$$\mathcal{J}_r F = r![s^r]F(s). \quad (1.1)$$

This equals $F^{(r)}(0)$ only when F is holomorphic there. A “first jet” means $[s]F(s)$ under this convention, not the finite part of $F'(s)$ under a separately chosen change of coordinate. The regulator coordinate s is fixed.

For an expansion at infinity, $[x^{-p}]f(x)$ means the coefficient in the Laurent series in x^{-1} . In particular, $[x^{-1}]$ is the negative of the complex-analytic residue at infinity when that terminology applies. This coefficient is computed from a finite truncation in all formulas below.

We use the standard Stieltjes normalization

$$\zeta(1+s, b) = \frac{1}{s} + \sum_{n \geq 0} \frac{(-1)^n}{n!} \gamma_n(b) s^n, \quad b > 0. \quad (1.2)$$

Thus $\gamma_0(b) = -\psi(b)$ and $\gamma_0(1) = \gamma$, Euler's constant. Spectral superscripts on ζ differentiate its first argument; the superscript on $\psi^{(k)}$ is an argument derivative. All logarithms of positive real quantities are real. The proofs also give local holomorphic shift continuations on any domain where compatible logarithms have been fixed.

1.4 What is not being claimed

An exact polynomial identity in the exponent variables is not an independence theorem for its special-function coefficients. A finite numerical residual is not a proof of equality. Neither a new Gamma-series coordinate nor a zeta-jet coordinate is claimed to be irreducible in a smaller arithmetic algebra. The ordinary proofs here have not been formalized in Lean or another proof assistant.

2 A transfer theorem for general discrete spectra

2.1 Hypotheses and a convergent expansion

Let (λ_ν) be a positive discrete spectrum, counted with multiplicity, with positive minimum λ_* and only finitely many indices below each fixed bound. Let $c_\nu \in \mathbb{C}$. Assume

$$\mathcal{D}(s) = \sum_{\nu} c_\nu \lambda_\nu^{-s} \quad (2.1)$$

converges absolutely for $\operatorname{Re} s > \sigma_a$, for some finite σ_a , and extends meromorphically to $\mathbb{C}$. The weights need not be positive. This is a hypothesis on $\mathcal{D}$, not a claimed property of every arithmetic sequence.

Let

$$P(x) = \sum_{k=0}^d p_k x^k, \quad a_j > -\lambda_*, \quad 1 \leq j \leq m, \quad U = \sum_{j=1}^m u_j.$$

Initially in $\operatorname{Re} U > \sigma_a + d$, define

$$\mathcal{Z}_P(\mathbf{u}; \mathbf{a}) = \sum_{\nu} c_\nu P(\lambda_\nu) \prod_{j=1}^m (\lambda_\nu + a_j)^{-u_j}. \quad (2.2)$$

The finite shifts do not change the absolute convergence condition at infinity. Introduce the germs

$$L_j(z) = \log(1 + a_j z), \quad L_{\mathbf{u}}(z) = \sum_j u_j L_j(z), \quad (2.3)$$

$$C_\ell(\mathbf{u}) = [z^\ell] e^{-L_{\mathbf{u}}(z)}. \quad (2.4)$$

Their coefficients are explicit polynomials:

$$C_\ell(\mathbf{u}) = \sum_{q_1 + \dots + q_m = \ell} \prod_{j=1}^m \frac{(-a_j)^{q_j} (u_j)_{q_j}}{q_j!}. \quad (2.5)$$

In particular, $C_0 = 1$, $C_\ell(0) = 0$ for $\ell > 0$, and $\deg C_\ell \leq \ell$.

Choose a finite head E consisting of all indices with $\lambda_\nu \leq R$, where $R > \max_j |a_j|$, and put

$$\mathcal{D}_E(s) = \mathcal{D}(s) - \sum_{\nu \in E} c_\nu \lambda_\nu^{-s}.$$

The finite subtraction is entire and leaves all principal parts unchanged. If the spectrum is finite, the desired conclusions hold with no poles and no infinite tail; hence suppose the tail is nonempty. Its minimum is denoted by M , so $M > \max |a_j|$.

Lemma 2.1 (Normally convergent tail expansion). *The meromorphic continuation of (2.2) is*

$$\mathcal{Z}_P(\mathbf{u}) = \mathcal{Z}_{P,E}(\mathbf{u}) + \sum_{k=0}^d p_k \sum_{\ell \geq 0} C_\ell(\mathbf{u}) \mathcal{D}_E(U + \ell - k), \quad (2.6)$$

where $\mathcal{Z}_{P,E}$ is the finite original head. On compact subsets away from the displayed polar divisors, the tail converges normally, as do all fixed mixed derivatives in $\mathbf{u}$.

Proof. Select ρ with $M^{-1} < \rho < (\max |a_j|)^{-1}$, interpreting the upper endpoint as infinity when every shift is zero. Cauchy's coefficient estimate on $|z| = \rho$ gives

$$|C_\ell(\mathbf{u})| \leq C_K \rho^{-\ell}$$

uniformly on every compact exponent set K . In a sufficiently far right half-plane, the absolutely convergent series for $\mathcal{D}_E$ gives, for bounded v ,

$$|\mathcal{D}_E(\ell + v)| \leq CM^{-\ell}.$$

Indeed fix $A > \sigma_a$ and majorize each term by $|c_\nu| \lambda_\nu^{-A} M^{-\ell+A-\operatorname{Re} v}$ for large ℓ . The resulting geometric factor is $(M\rho)^{-\ell} < 1$. Fixed derivatives of $\mathcal{D}_E$ add powers of $\log \lambda_\nu$; choosing a slightly larger absolute-convergence exponent absorbs them. Derivatives of C_ℓ are controlled by Cauchy estimates on a larger compact exponent set.

Expanding every shifted factor by its binomial series proves (2.6) in the initial convergence region, where these estimates justify interchanging the sums. The normally convergent right side then supplies the meromorphic continuation. Only finitely many small ℓ can contribute a pole on a fixed compact exponent set, because the other arguments lie in an absolute-convergence half-plane. $\square$

2.2 The complete local polar numerator

Let $\mathcal{P}$ be the set of integer poles p of $\mathcal{D}$ with $p \geq -d$. It is finite: there are no poles to the right of an absolute-convergence half-plane. Write the principal part at each such pole as

$$\mathcal{D}(p+s) = \sum_{h=1}^{H_p} \frac{\kappa_{p,h}}{s^h} + \text{a holomorphic function of } s. \quad (2.7)$$

The highest coefficient κ_{p,H_p} is nonzero. Define

$$\mathcal{R}_p(\mathbf{u}) = \sum_{\substack{0 \leq k \leq d \\ k+p \geq 0}} p_k C_{k+p}(\mathbf{u}). \quad (2.8)$$

A pole with $\mathcal{R}_p = 0$ contributes nothing. Noninteger poles do not pass through the origin of exponent space.

Theorem 2.2 (Finite-principal-part transfer). *In a neighborhood of $\mathbf{u} = 0$, the exact germ is*

$$\mathcal{Z}_P(\mathbf{u}) = \mathcal{H}_P(\mathbf{u}) + \sum_{p \in \mathcal{P}} \sum_{h=1}^{H_p} \frac{\kappa_{p,h} \mathcal{R}_p(\mathbf{u})}{U^h}, \quad (2.9)$$

where $\mathcal{H}_P$ is holomorphic. The numerator $\mathcal{R}_p$ is a polynomial of degree at most $d + p$. If $p > 0$ then $\mathcal{R}_p(0) = 0$. Neither the principal coefficients nor the numerators depend on the finite head used in Lemma 2.1.

Proof. A term $\mathcal{D}_E(U + \ell - k)$ has a pole through $U = 0$ precisely when $p = \ell - k$ is an integer pole of $\mathcal{D}$. For a fixed such p , its coefficient is $\sum_k p_k C_{k+p} = \mathcal{R}_p$. Subtract the finitely many principal parts (2.7) from these terms. The remaining small- ℓ terms are holomorphic near the origin, and the large- ℓ tail is holomorphic there by Lemma 2.1. Choose the neighborhood small enough to avoid every other pole of the finitely many remaining small- ℓ factors. This proves (2.9). The degree and constant-term assertions follow from (2.5). Finally, altering a finite head changes $\mathcal{D}_E$ by an entire function, which cannot change any principal coefficient. Formula (2.8) uses only P and the shifts. $\square$

A zero numerator at $\mathbf{u} = 0$ does not imply a holomorphic ray restriction when $H_p > 1$. It removes at most one order of the pole without additional cancellation. Conversely, if $\mathcal{D}$ has no relevant integer poles, then $\mathcal{Z}_P$ is jointly holomorphic at the origin even if $\mathcal{D}$ has poles elsewhere. These are local statements at the specified spectral point.

3 All-order local identities and finite differences

Normalize the exponent vector by $\sum_j \alpha_j = 1$, and set

$$\mathcal{J}_r(\boldsymbol{\alpha}) = r! [s^r] \mathcal{Z}_P(s\boldsymbol{\alpha}), \quad r \geq 0. \quad (3.1)$$

Let $F_{[q]}$ denote the homogeneous Taylor part of degree q of a holomorphic germ or polynomial F . Put $H = \max_{p \in \mathcal{P}} H_p$, with $H = 0$ if $\mathcal{P}$ is empty.

Theorem 3.1 (Polynomiality of every normalized Laurent jet). *On the affine hyperplane $\sum \alpha_j = 1$,*

$$\mathcal{J}_r(\boldsymbol{\alpha}) = r! \mathcal{H}_{P,[r]}(\boldsymbol{\alpha}) + r! \sum_{p,h} \kappa_{p,h} \mathcal{R}_{p,[r+h]}(\boldsymbol{\alpha}), \quad (3.2)$$

$$\mathcal{R}_{p,[q]}(\boldsymbol{\alpha}) = \frac{(-1)^q}{q!} [x^{-p}] P(x) \left(\sum_j \alpha_j \log(1 + a_j/x) \right)^q. \quad (3.3)$$

Consequently $\deg \mathcal{J}_r \leq r + H$. A local term of degree $r + h$ can occur only when $r + h \leq d + p$. Every term beyond degree r is determined by the finite principal parts of $\mathcal{D}$.

Proof. On the normalized ray $U = s$. Extracting the coefficient of s^r in (2.9) gives (3.2). The homogeneous part of degree q of $e^{-L_{\mathbf{u}}(z)}$ is $(-1)^q L_{\mathbf{u}}(z)^q / q!$. Using (2.8) and then $z = x^{-1}$ proves (3.3). Every factor of the logarithmic sum starts with x^{-1} , proving the sharper degree cutoff. $\square$

For a tangent vector $\mathbf{v}$ with $\sum_j v_j = 0$, write

$$\Delta_{\mathbf{v}} f(\boldsymbol{\alpha}) = f(\boldsymbol{\alpha} + \mathbf{v}) - f(\boldsymbol{\alpha}), \quad V_{\mathbf{v}}(x) = \sum_j v_j \log(x + a_j).$$

The $\log x$ terms cancel, so $V_{\mathbf{v}}(x) = O(x^{-1})$.

Theorem 3.2 (Top local polarization). *Suppose $H \geq 1$ and let $q = r + H$. For any q tangent vectors,*

$$\begin{aligned} \Delta_{v_1} \cdots \Delta_{v_q} \mathcal{J}_r(\alpha) \\ = (-1)^q r! \sum_{p: H_p = H} \kappa_{p,H} [x^{-p}] P(x) \prod_{i=1}^q V_{v_i}(x). \end{aligned} \quad (3.4)$$

The same expression equals the corresponding mixed directional derivative. Every mixed difference of order $q + 1$ vanishes.

Proof. The q differentiations annihilate $\mathcal{H}_{P,[r]}$ and every local part of degree below q . Polarizing the remaining q th power in (3.3) supplies exactly $q! \prod_i V_{v_i}$. For a polynomial of degree at most q , a mixed q th finite difference is its constant mixed q th derivative; integrate that derivative over the unit q -cube to see the equality directly. Differences of the next order vanish by the degree bound. $\square$

The highest pole alone controls (3.4); it does not control all lower local differences. The following formula retains every layer.

Theorem 3.3 (All finite local differences). *Let $q > r$, and put $L_\alpha(x) = \sum_j \alpha_j \log(1 + a_j/x)$. Then*

$$\begin{aligned} \Delta_{v_1} \cdots \Delta_{v_q} \mathcal{J}_r(\alpha) \\ = r! \sum_{p,h} \kappa_{p,h} [x^{-p}] P(x) [t^{r+h}] e^{-tL_\alpha(x)} \prod_{i=1}^q (e^{-tV_{v_i}(x)} - 1). \end{aligned} \quad (3.5)$$

The right side is a finite algebraic computation. Terms with $q > r + h$ vanish automatically.

Proof. The differences kill the degree- r holomorphic part. Apply a difference to e^{-tL_α} : it multiplies the exponential by $e^{-tV_v} - 1$. Repeating and extracting t^{r+h} in (3.2) gives the assertion. Each of the q factors is divisible by t , proving the last statement. Coefficients in x^{-1} need only be retained through the largest integer $d + p$ occurring. $\square$

3.1 Normalization, scaling, and the meaning of locality

For arbitrary weights $\mathbf{w}$ with $W = \sum_j w_j \neq 0$,

$$r! [s^r] \mathcal{Z}_P(s\mathbf{w}) = W^r \mathcal{J}_r(\mathbf{w}/W). \quad (3.6)$$

This remains true for Laurent series: substitute $t = Ws$ before extracting the coefficient. The excluded hyperplane $W = 0$ cannot be restored by blindly taking a limit in these formulas; it lies in the polar divisor and needs a separately specified restriction.

“Local” here has a precise meaning: the expression depends on finitely many principal coefficients of the base Dirichlet series and finitely many terms of the large- x logarithmic expansion. It is not a claim that the individual jets have elementary evaluations. Finite-head changes affect only $\mathcal{H}_P$, so any identity obtained by more than r tangent differences is insensitive to finitely many altered spectral terms.

4 Periodic weights and several distinct arithmetic poles

4.1 A complete quasi-polynomial specialization

Let $q \geq 1$ and let $a_0, \dots, a_K$ be periodic functions on the positive integers with period q . Define

$$c_n = \sum_{d=0}^K n^d a_d(n), \quad \mu_d = \frac{1}{q} \sum_{j=1}^q a_d(j).$$

Residue-class splitting in the absolute-convergence region gives

$$\mathcal{D}(s) = \sum_{d=0}^K q^{d-s} \sum_{j=1}^q a_d(j) \zeta(s-d, j/q). \quad (4.1)$$

This finite formula proves the continuation hypotheses of Theorem 2.2. Its only possible poles are simple, at $p = d + 1$, with residues μ_d .

Corollary 4.1 (Only the mean polynomial contributes local defects). *For quasi-polynomial multiplicities and any polynomial P , all normalized order- r jets have degree at most $r + 1$. For $r + 1$ tangent vectors,*

$$\begin{aligned} & \Delta_{v_1} \cdots \Delta_{v_{r+1}} \mathcal{J}_r(\alpha) \\ &= (-1)^{r+1} r! [x^{-1}] P(x) \left(\sum_{d=0}^K \mu_d x^d \right) \prod_i V_{v_i}(x). \end{aligned} \quad (4.2)$$

If every $\mu_d = 0$, the degree is at most r and the displayed difference vanishes. More generally, replacing c_n by its mean polynomial changes only a degree-at-most- r part of the normalized jet.

Proof. Use the residues in (4.1) in (3.4) and observe $[x^{-(d+1)}]f(x) = [x^{-1}]x^d f(x)$. Subtracting the mean polynomial leaves a base Dirichlet series whose possible poles have all been canceled, so its shifted germ is holomorphic at the exponent origin. Theorem 3.1 then gives degree at most r . $\square$

For example, a nontrivial additive character $a(n) = e^{2\pi i k n/q}$ has zero mean. Every normalized order- r jet for this periodically weighted spectrum, even after multiplication by $n^d P(n)$, has degree at most r . There is no first-derivative multiplicative anomaly for such a pure zero-mean periodic component. This does not say that its individual jets vanish.

The finite parts themselves retain the periodic data. For one degree d , the constant coefficient at $s = d + 1$ of the corresponding term in (4.1) is

$$\frac{1}{q} \sum_{j=1}^q a_d(j) \gamma_0(j/q) - \mu_d \log q.$$

Other degrees contribute holomorphic values there. Higher coefficients follow by multiplying $e^{-s \log q}$ by (1.2). Thus the mean law is a statement about the local defect, not the entire answer.

4.2 A two-pole example from the sum-of-divisors function

Let $\sigma_1(n) = \sum_{d|n} d$. The elementary product of two absolutely convergent Dirichlet series proves

$$\sum_{n \geq 1} \frac{\sigma_1(n)}{n^s} = \zeta(s) \zeta(s-1), \quad \operatorname{Re} s > 2. \quad (4.3)$$

The two poles are simple, with

$$\kappa_{1,1} = \zeta(0) = -\frac{1}{2}, \quad \kappa_{2,1} = \zeta(2).$$

The coexistence of distinct poles is genuinely different from inserting a power of $\log n$ into one polynomially weighted spectrum.

For $P = 1$, denote the continued multi-shift series by Z_w , and the single-shift series by F_a . Both are holomorphic at the spectral origin, since all relevant numerators vanish there and the poles are simple. Set

$$\delta_w = \sum_j w_j F'_{a_j}(0) - Z'_w(0), \quad W \neq 0.$$

Proposition 4.2 (An exact arithmetic anomaly). *For $a_j > -1$,*

$$\delta_w = \frac{\zeta(2)}{2W} \sum_{i < j} w_i w_j (a_i - a_j)^2. \quad (4.4)$$

Proof. Normalize the weights. For $r = 1$, the nonlocal part is linear and cancels against interpolation at the coordinate vertices. The pole at 1 gives $[x^{-1}]L_\alpha^2 = 0$. The pole at 2 gives $\zeta(2)(\sum_j \alpha_j a_j)^2/2$. Therefore the difference between vertex interpolation and the jet is

$$\frac{\zeta(2)}{2} \left\{ \sum_j \alpha_j a_j^2 - \left(\sum_j \alpha_j a_j \right)^2 \right\} = \frac{\zeta(2)}{2} \sum_{i < j} \alpha_i \alpha_j (a_i - a_j)^2.$$

Multiply by W using (3.6). The small-shift argument extends to all stated shifts by the finite-head theorem. $\square$

For two unit weights and shifts a, b , the anomaly is $\zeta(2)(a - b)^2/4$. This is a concrete arithmetic specialization of the classical simple-pole mechanism, not a claim to have discovered that mechanism. In contrast, the next sections concern base functions with higher poles, for which $F'_a(0)$ is typically not defined at all.

5 Multiplicative Hurwitz lattices and all Stieltjes layers

5.1 An unconditional higher-pole family

Fix $h \geq 1$ and $b_1, \dots, b_h > 0$. Let

$$\lambda_{\mathbf{n}} = \prod_{j=1}^h (n_j + b_j), \quad \mathbf{n} \in \mathbb{N}_0^h, \quad \lambda_* = \prod_j b_j. \quad (5.1)$$

This spectrum is discrete with finite heads. Multiplying absolutely convergent series gives

$$\mathcal{D}(s) = \sum_{\mathbf{n}} \lambda_{\mathbf{n}}^{-s} = \prod_{j=1}^h \zeta(s, b_j), \quad \operatorname{Re} s > 1. \quad (5.2)$$

Thus there is just one pole, at $s = 1$, of order exactly h , and its leading coefficient is one. When all $b_j = 1$, collecting equal integer products gives

$$\mathcal{D}(s) = \sum_{N \geq 1} \frac{d_h(N)}{N^s} = \zeta(s)^h, \quad d_h(N) = \#\{(n_1, \dots, n_h) \in \mathbb{N}^h : n_1 \cdots n_h = N\}.$$

These are nonpolynomial arithmetic multiplicities. In particular, $d_2(N)$ is the ordinary number-of-divisors function.

Write

$$\mathcal{D}(1+s) = \sum_{v=-h}^{\infty} d_v s^v. \quad (5.3)$$

The notation d_v here is a Laurent coefficient, distinguished from the arithmetic function $d_h(N)$ by its argument. Define

$$A_j(s) = 1 + \sum_{n \geq 0} \frac{(-1)^n \gamma_n(b_j)}{n!} s^{n+1}.$$

Then the exact finite recipe is

$$d_v = [s^{v+h}] \prod_{j=1}^h A_j(s). \quad (5.4)$$

For any fixed v this uses only finitely many Stieltjes constants. In particular,

$$\begin{aligned} d_{-h} &= 1, \\ d_{-(h-1)} &= \sum_j \gamma_0(b_j), \\ d_{-(h-2)} &= \sum_{i < j} \gamma_0(b_i) \gamma_0(b_j) - \sum_j \gamma_1(b_j). \end{aligned} \quad (5.5)$$

For two parameters b, c , the first regular coefficients are

$$d_0 = \gamma_0(b) \gamma_0(c) - \gamma_1(b) - \gamma_1(c), \quad (5.6)$$

$$d_1 = \frac{\gamma_2(b) + \gamma_2(c)}{2} - \gamma_0(b) \gamma_1(c) - \gamma_1(b) \gamma_0(c), \quad (5.7)$$

$$d_2 = -\frac{\gamma_3(b) + \gamma_3(c)}{6} + \frac{\gamma_0(b) \gamma_2(c) + \gamma_2(b) \gamma_0(c)}{2} + \gamma_1(b) \gamma_1(c). \quad (5.8)$$

These follow by multiplying (1.2); no regularized product of divergent constants is involved.

5.2 Every local term, not only the highest pole

For the lattice (5.1), Theorem 2.2 becomes

$$\mathcal{Z}_P(\mathbf{u}) = \mathcal{H}_P(\mathbf{u}) + \mathcal{R}_1(\mathbf{u}) \sum_{j=1}^h \frac{d_{-j}}{U^j}, \quad \mathcal{R}_1 = \sum_{k=0}^d p_k C_{k+1}. \quad (5.9)$$

Consequently the pole coefficients are independent of the auxiliary finite head but depend nontrivially on the Hurwitz parameters. The coefficient $d_{-h} = 1$ makes the very highest difference independent of those parameters:

$$\Delta_{v_1} \cdots \Delta_{v_{r+h}} \mathcal{J}_r = (-1)^{r+h} r! [x^{-1}] P(x) \prod_i V_{v_i}(x). \quad (5.10)$$

If P has degree less than $r+h-1$, this particular difference vanishes. If P has degree exactly $r+h-1$ with leading coefficient p , it equals

$$(-1)^{r+h} r! p \prod_{i=1}^{r+h} \left(\sum_j v_{ij} a_j \right). \quad (5.11)$$

Thus the degree bound is attained whenever this product is nonzero. Lower differences generally retain the Stieltjes coefficients in (5.5); replacing the whole principal part by its leading term would give incorrect lower-order formulas.

5.3 A normally convergent expression for the individual jets

Put

$$A_{t,\ell}(\boldsymbol{\alpha}) = [z^\ell] \left(\sum_j \alpha_j \log(1 + a_j z) \right)^t, \quad A_{0,0} = 1, \quad A_{0,\ell} = 0 \quad (\ell > 0).$$

The following formula makes the nonlocal coordinates explicit as well.

Theorem 5.1 (All individual Hurwitz-product jets). *Suppose $\max_j |a_j| < \lambda_*$. For every $r \geq 0$,*

$$\begin{aligned} \mathcal{J}_r(\boldsymbol{\alpha}) = \sum_{k=0}^d p_k \Bigg\{ & \mathcal{D}^{(r)}(-k) + \sum_{\substack{\ell \geq 1 \\ \ell \neq k+1}} \sum_{t=1}^r (-1)^t \binom{r}{t} A_{t,\ell} \mathcal{D}^{(r-t)}(\ell - k) \\ & + r! \sum_{t=1}^{r+h} \frac{(-1)^t}{t!} A_{t,k+1} d_{r-t} \Bigg\}. \end{aligned} \quad (5.12)$$

Every infinite sum converges normally on smaller compact shift and exponent sets. For arbitrary allowed real shifts, the same construction uses a finite original head and its entire subtraction from $\mathcal{D}$.

Proof. For the stated small shifts, take an empty head in (2.6) and expand

$$C_\ell(s\boldsymbol{\alpha}) = \sum_{t \geq 0} \frac{(-1)^t}{t!} A_{t,\ell} s^t.$$

At $\ell = 0$, the contribution is $\mathcal{D}^{(r)}(-k)$. At nonresonant $\ell \geq 1$, multiply by the Taylor series of $\mathcal{D}(s + \ell - k)$. At the unique resonant value $\ell = k + 1$, multiply instead by (5.3). The coefficient of s^r receives terms through $t = r + h$, exactly as displayed. Lemma 2.1 justifies every coefficient extraction and proves the finite-head extension. $\square$

The last sum is finite, but its upper endpoint is essential. Truncating $C_{k+1}(s\boldsymbol{\alpha})$ at degree r before multiplying by the pole loses all terms responsible for the local identities.

5.4 A direct obstruction to using ordinary derivatives

For one shifted factor and $P = 1$, write

$$F_a(s) = \sum_n (\lambda_n + a)^{-s}.$$

Since $C_1(s) = -as$, the only polar contribution at zero is

$$\text{pp}_{s=0} F_a(s) = -a \sum_{j=2}^h d_{-j} s^{1-j}. \quad (5.13)$$

For $h = 2$, in particular,

$$\text{Res}_{s=0} F_a(s) = -a, \quad [s^0] F_a(s) = \mathcal{D}(0) - a\{\gamma_0(b) + \gamma_0(c)\}. \quad (5.14)$$

Thus $F'_a(0)$ does not exist for $a \neq 0$. This is a proved warning against an overextension of the ordinary determinant convention, not an error attributed to the earlier report, which already distinguished Laurent coefficients for logarithmic multiplicities.

6 Exact alternating identities for arithmetic Laurent jets

Let $m \geq 2$,

$$N = \binom{m}{2}, \quad K_m = \prod_{k=0}^{m-1} k!, \quad \Delta(\mathbf{x}) = \prod_{i < j} (x_j - x_i).$$

We use $(\sigma \mathbf{w})_j = w_{\sigma(j)}$ and $W = \sum_j w_j \neq 0$. For the multiplicative Hurwitz lattice of order h assume $h \leq N$, and set $r = N - h$.

Theorem 6.1 (Higher-pole spectral alternant). *For every polynomial P and allowed shifts,*

$$\begin{aligned} & \sum_{\sigma \in S_m} \operatorname{sgn}(\sigma) r! [s^r] \mathcal{Z}_P(s\sigma \mathbf{w}) \\ &= \frac{(-1)^N r!}{W^h K_m} \Delta(\mathbf{w}) [x^{-1}] P(x) \prod_{i < j} \log \frac{x + a_j}{x + a_i}. \end{aligned} \quad (6.1)$$

If P has degree $N - 1$ and leading coefficient p , this reduces to

$$\boxed{\sum_{\sigma} \operatorname{sgn}(\sigma) r! [s^r] \mathcal{Z}_P(s\sigma \mathbf{w}) = \frac{(-1)^N r! p}{W^h K_m} \Delta(\mathbf{w}) \Delta(\mathbf{a}).} \quad (6.2)$$

For degree below $N - 1$, the alternating sum is zero.

Proof. An alternating polynomial in m variables is divisible by their Vandermonde, whose degree is N . In (3.2), alternation therefore kills the holomorphic term of degree $r < N$ and every local term of degree $r + j < N$. Only the top principal coefficient $d_{-h} = 1$ remains. The elementary identity

$$\sum_{\sigma} \operatorname{sgn}(\sigma) \left(\sum_j \alpha_{\sigma(j)} y_j \right)^N = \frac{N!}{K_m} \Delta(\boldsymbol{\alpha}) \Delta(\mathbf{y}) \quad (6.3)$$

follows by expanding the power. Terms with a repeated exponent vanish; the only distinct nonnegative exponents summing to N are $0, 1, \dots, m - 1$. Their two determinants give the Vandermondes with the orientation displayed. Substituting $y_j = \log(1 + a_j/x)$ in (3.3) proves the normalized identity. Equation (3.6) and $\Delta(\mathbf{w}/W) = W^{-N} \Delta(\mathbf{w})$ produce $W^{r-N} = W^{-h}$. Finally,

$$\prod_{i < j} \log \frac{x + a_j}{x + a_i} = \Delta(\mathbf{a}) x^{-N} + O(x^{-N-1}),$$

which gives the last two assertions. $\square$

6.1 A six-term first-jet identity for the divisor function

For a monic quadratic P define

$$T_P(a, b) = [s] \sum_{n \geq 1} d_2(n) P(n) ((n + a)^2 (n + b))^{-s}, \quad a, b > -1, \quad (6.4)$$

by meromorphic continuation and the fixed Laurent convention.

Corollary 6.2 (Six first jets collapse to an algebraic number).

$$\begin{aligned} & T_P(a, b) + T_P(b, c) + T_P(c, a) \\ & - T_P(b, a) - T_P(c, b) - T_P(a, c) = \frac{(a - b)(b - c)(c - a)}{9}. \end{aligned} \quad (6.5)$$

The lower two coefficients of P do not affect the answer. For a quadratic leading coefficient p , multiply the right side by p ; for degree at most one the answer is zero.

Proof. Apply Theorem 6.1 with $m = 3$, $h = 2$, $r = 1$, $\mathbf{w} = (2, 1, 0)$, $W = 3$, $K_3 = 2$, and $\Delta(\mathbf{w}) = -2$. Reading the six permutations in (6.2) gives precisely (6.5). $\square$

A compact direct certificate. For the six permutations of $(2, 1, 0)$ let $\mathcal{A}$ denote signed summation. Since $C_\ell(s\mathbf{w})$ is polynomial in the weights, its alternating part is divisible by s^3 . It vanishes identically for $\ell < 3$. At $\ell = 3$, equation (6.3) gives

$$\mathcal{A}C_3(s\mathbf{w}) = s^3\Delta(a, b, c).$$

For $P(x) = x^2$, only $\ell = 3$ can contribute to the first Laurent coefficient: all other nonzero alternating terms multiply a function holomorphic at $s = 0$. The resonant factor is $\zeta(1 + 3s)^2 = 1/(9s^2) + O(s^{-1})$. Hence its first coefficient is $\Delta(a, b, c)/9$. The lower coefficients of P resonate only at $\ell < 3$ and make no contribution. This is a finite symbolic certificate of the corollary independent of any numerical spectral evaluation.

For $(a, b, c) = (1, 2, 3)$ the value is $2/9$. Replacing $d_2(n)$ by the three-factor divisor multiplicity $d_3(n)$ moves the same identity down one more spectral order: with

$$U_P(a, b) = [s^0] \sum_{n \geq 1} d_3(n) P(n) ((n+a)^2(n+b))^{-s},$$

the same signed six-term combination equals

$$\frac{(a-b)(b-c)(c-a)}{27}. \quad (6.6)$$

It is an identity between constant Laurent coefficients, not between values of functions at their poles.

6.2 A hierarchy with independent Hurwitz parameters

The identities (6.5) and (6.6) hold unchanged when the integer divisor spectrum is replaced by arbitrary multiplicative Hurwitz lattices of two and three factors, respectively. Although each individual Laurent jet depends on all the b_j , the alternating combination does not. The reason is exactly $d_{-h} = 1$, not a cancellation suggested by numerical fitting.

For $m = 4$, $N = 6$ and $K_4 = 12$. With $h = 2$, a signed sum of twenty-four fourth jets for a degree-five polynomial with leading coefficient p is

$$\frac{2p}{W^2} \Delta(\mathbf{w}) \Delta(\mathbf{a}).$$

With $h = 3$, the corresponding third-jet sum is $p\Delta(\mathbf{w})\Delta(\mathbf{a})/(2W^3)$. This makes visible the general shift in spectral order: the same Vandermonde identity occurs at order $N - h$ when the base pole has order h . These are exact consequences of the higher-pole germ; applying a simple-pole determinant formula at an undefined derivative would not establish them.

7 Gamma reciprocity, polygamma sums, and a normalized primitive

7.1 A convergent nonlinear part of the first Laurent jet

For the multiplicative Hurwitz lattice define

$$E_b(a) = \sum_n \left\{ \frac{a}{\lambda_n} - \log \left(1 + \frac{a}{\lambda_n} \right) \right\}, \quad a > -\lambda_*. \quad (7.1)$$

Every brace must be kept grouped. Its tail is $O(\lambda_n^{-2})$ on compact shift sets, and $\sum \lambda_n^{-2} = \prod_j \zeta(2, b_j) < \infty$. Thus this is an ordinary, locally normally convergent series, not a finite-part sum.

Proposition 7.1 (First-jet decomposition). *For F_a of Section 5,*

$$\boxed{\mathcal{J}_1 F_a = \mathcal{D}'(0) - ad_0 + E_b(a)}. \quad (7.2)$$

Proof. For $|a| < \lambda_*$, apply (5.12) with $P = 1$, one exponent, and $r = 1$. The resonant term is $-ad_0$. For $\ell \geq 2$, $s_\ell/\ell! = 1/\ell$, so the remaining sum is

$$\sum_{\ell \geq 2} \frac{(-a)^\ell}{\ell} \mathcal{D}(\ell) = \sum_n \{a/\lambda_n - \log(1 + a/\lambda_n)\}.$$

Absolute convergence justifies the interchange. Both sides continue locally holomorphically throughout the slit shift plane $\mathbb{C} \setminus (-\infty, -\lambda_*]$, using finite heads on the spectral side and the grouped series on the other side. The identity theorem, or continuation along the real interval, gives every stated a . $\square$

The entire principal part at the spectral origin is affine in a by (5.13). All nonlinearity of the first Laurent coefficient is therefore the convergent function E_b .

7.2 Two independent Gamma representations of the same function

For two parameters write $E_{b,c} = E_{(b,c)}$.

Theorem 7.2 (Centered Gamma reciprocity). *For $b, c > 0$ and $a > -bc$,*

$$\begin{aligned} E_{b,c}(a) &= \sum_{n=0}^{\infty} \left\{ \log \Gamma\left(c + \frac{a}{n+b}\right) - \log \Gamma(c) - \frac{a\psi(c)}{n+b} \right\} \\ &= \sum_{m=0}^{\infty} \left\{ \log \Gamma\left(b + \frac{a}{m+c}\right) - \log \Gamma(b) - \frac{a\psi(b)}{m+c} \right\}. \end{aligned} \quad (7.3)$$

Both series converge absolutely and locally uniformly in a . On $|a| < bc$ they also equal

$$E_{b,c}(a) = \sum_{\ell \geq 2} \frac{(-a)^\ell}{\ell} \zeta(\ell, b) \zeta(\ell, c). \quad (7.4)$$

Proof. The classical centered Gamma product gives, for $c > 0$ and $t > -c$,

$$\log \Gamma(c+t) - \log \Gamma(c) - t\psi(c) = \sum_{m \geq 0} \left\{ \frac{t}{m+c} - \log \left(1 + \frac{t}{m+c} \right) \right\}. \quad (7.5)$$

One may derive it without choosing a product normalization: both sides vanish with their first derivative at $t = 0$, and their second derivatives are the same convergent trigamma series. Set $t = a/(n+b)$ and sum over n . All double summands are absolutely summable, since their tails are bounded by a constant times $(n+b)^{-2}(m+c)^{-2}$ and their finitely many small factors stay away from zero on compact allowed shift sets. Fubini's theorem gives $E_{b,c}$. Interchanging m and n proves the second representation. Expanding the logarithm under $|a| < bc$ proves (7.4). $\square$

For $b = c = 1$, the first Laurent coefficient becomes the explicit Gamma–Stieltjes identity

$$\mathcal{J}_1 F_a = \frac{1}{2} \log(2\pi) - a(\gamma^2 - 2\gamma_1) + \sum_{n \geq 1} \left\{ \log \Gamma(1 + a/n) + \frac{\gamma a}{n} \right\}. \quad (7.6)$$

Here $\mathcal{D}'(0) = 2\zeta(0)\zeta'(0) = \frac{1}{2} \log(2\pi)$ and $d_0 = \gamma^2 - 2\gamma_1$. At the same time,

$$\text{Res}_{s=0} F_a = -a, \quad [s^0]F_a = \frac{1}{4} - 2\gamma a.$$

The Gamma series in (7.6) cannot make that nonzero pole disappear. It evaluates the correctly specified Laurent coefficient.

7.3 Reciprocal digamma and all polygamma derivatives

Corollary 7.3 (All argument derivatives). *For $b, c > 0$, $a > -bc$,*

$$\begin{aligned} E'_{b,c}(a) &= \sum_{n \geq 0} \frac{\psi(c + a/(n+b)) - \psi(c)}{n+b} \\ &= \sum_{m \geq 0} \frac{\psi(b + a/(m+c)) - \psi(b)}{m+c}. \end{aligned} \quad (7.7)$$

For every integer $r \geq 2$,

$$\begin{aligned} E_{b,c}^{(r)}(a) &= \sum_{n \geq 0} \frac{\psi^{(r-1)}(c + a/(n+b))}{(n+b)^r} \\ &= \sum_{m \geq 0} \frac{\psi^{(r-1)}(b + a/(m+c))}{(m+c)^r} \\ &= (-1)^r (r-1)! \sum_{n,m \geq 0} ((n+b)(m+c) + a)^{-r}. \end{aligned} \quad (7.8)$$

At $a = 0$ this is $(-1)^r (r-1)! \zeta(r, b) \zeta(r, c)$.

Proof. Differentiate the grouped normally convergent series on compact subsets of the allowed shift domain. The r th derivative for $r \geq 2$ of one grouped summand in (7.1) is $(-1)^r (r-1)! (\lambda + a)^{-r}$. The Gamma representation gives the first two forms. The conventional polygamma series, or differentiation of (7.5), gives the last one. At $a = 0$ absolute convergence factors the double series. $\square$

For more than two Hurwitz factors, summing over one factor gives an analogous centered Gamma series over the remaining $(h-1)$ indices. The choice of the factor summed first is arbitrary, producing h compatible Gamma representations. This is a constructive representation, not a reduction claim into finitely many Gamma values.

7.4 An anchored primitive in Hurwitz-zeta derivatives

Define the classical primitive

$$K(z) = \zeta'(-1, z) + \frac{z - z^2}{2} + \frac{z}{2} \log(2\pi), \quad z > 0. \quad (7.9)$$

The identities $\partial_z \zeta(s, z) = -s\zeta(s+1, z)$ and $\zeta'(0, z) = \log \Gamma(z) - \frac{1}{2} \log(2\pi)$ imply $K'(z) = \log \Gamma(z)$ [3]. This agrees with the primitive already used in the manuscript [1].

Theorem 7.4 (Reciprocal normalized primitive). *Let $I_{b,c}(a) = \int_0^a E_{b,c}(t) dt$, with oriented integration when $a < 0$. Then*

$$I_{b,c}(a) = \sum_{n \geq 0} \left\{ (n+b) \left[K\left(c + \frac{a}{n+b}\right) - K(c) \right] - a \log \Gamma(c) - \frac{a^2 \psi(c)}{2(n+b)} \right\}. \quad (7.10)$$

The same expression with b and c exchanged is equal to it. Equivalently,

$$I_{b,c}(a) = \sum_{n,m \geq 0} \left\{ \frac{a^2}{2\lambda_{n,m}} - (\lambda_{n,m} + a) \log \left(1 + \frac{a}{\lambda_{n,m}} \right) + a \right\}, \quad (7.11)$$

where $\lambda_{n,m} = (n+b)(m+c)$. For $|a| < bc$,

$$I_{b,c}(a) = \sum_{\ell \geq 2} \frac{(-1)^\ell a^{\ell+1}}{\ell(\ell+1)} \zeta(\ell, b) \zeta(\ell, c). \quad (7.12)$$

Proof. Integrate the grouped summands of Theorem 7.2 along the compact segment from 0 to a . Uniform convergence permits the interchange. Substitution $u = c + t/(n+b)$ gives the K term, and the two subtractions integrate to the other two terms in (7.10). Integration of (7.1) gives (7.11). Its grouped tail is $O(\lambda_{n,m}^{-2})$. Both forms vanish at $a = 0$, fixing the integration constant. Integrating (7.4) proves (7.12). $\square$

In (7.10), the three large pieces must not be summed separately. The cancellations are part of the identity's definition. This is the same normalization issue that occurs in negative polygamma, but here both the base point and every convergent subtraction are explicit.

8 Generalized harmonic numbers in every single-shift jet

Let $e_j(q)$ be the elementary symmetric polynomial of degree j in $1, 1/2, \dots, 1/q$, with $e_0(q) = 1$ and $e_j(q) = 0$ for $j > q$. For $q = 0$ use the empty alphabet. Define $H_q^{(j)} = \sum_{n=1}^q n^{-j}$ and $H_q = H_q^{(1)}$. The finite product gives

$$\begin{aligned} \frac{(s)_\ell}{\ell!} &= \frac{s}{\ell} \prod_{j=1}^{\ell-1} (1 + s/j) = \frac{s}{\ell} \sum_{j=0}^{\ell-1} e_j(\ell-1) s^j, \\ \sum_{j \geq 0} e_j(q) s^j &= \exp \left(\sum_{k \geq 1} \frac{(-1)^{k+1}}{k} H_q^{(k)} s^k \right). \end{aligned} \quad (8.1)$$

The second equality is understood as a power-series identity near zero; its exponential simplifies to the finite product. In particular,

$$e_1(q) = H_q, \quad e_2(q) = \frac{1}{2} \{H_q^2 - H_q^{(2)}\}, \quad e_3(q) = \frac{1}{6} \{H_q^3 - 3H_q H_q^{(2)} + 2H_q^{(3)}\}.$$

These are the familiar harmonic/Stirling coordinates already present in the manuscript's derivative tower. Their role here is to control arithmetic spectral finite parts, not just parameter derivatives of a single Hurwitz function.

Theorem 8.1 (All single-shift Laurent jets). *For the multiplicative Hurwitz lattice, $|a| < \lambda_*$, and $r \geq 1$,*

$$\begin{aligned} \mathcal{J}_r F_a &= \mathcal{D}^{(r)}(0) - ar!d_{r-1} \\ &+ \sum_{\ell \geq 2} \frac{(-a)^\ell}{\ell} \sum_{t=1}^{\min(r,\ell)} \frac{r!}{(r-t)!} e_{t-1}(\ell-1) \mathcal{D}^{(r-t)}(\ell). \end{aligned} \quad (8.2)$$

The constant Laurent coefficient is $\mathcal{J}_0 F_a = \mathcal{D}(0) - ad_{-1}$. All spectral derivatives of $\mathcal{D}$ are finite sums of products of Hurwitz-zeta derivatives by Leibniz's rule. Thus (8.2) uses only specified zeta jets, generalized harmonic numbers, and finite Stieltjes polynomials.

Proof. The binomial expansion is

$$F_a(s) = \sum_{\ell \geq 0} \frac{(-a)^\ell (s)_\ell}{\ell!} \mathcal{D}(s + \ell).$$

The $\ell = 0$ term contributes $\mathcal{D}^{(r)}(0)$. The resonant term $\ell = 1$ is $-as\mathcal{D}(1+s)$ and contributes $-ar!d_{r-1}$, including $r = 0$. For $\ell \geq 2$, substitute (8.1) and multiply by the ordinary Taylor series of $\mathcal{D}(\ell + s)$. Theorem 5.1 justifies the extraction and convergence. $\square$

The first genuinely higher rows are worth displaying without compressed symmetric-function notation:

$$\mathcal{J}_2 F_a = \mathcal{D}''(0) - 2ad_1 + 2 \sum_{\ell \geq 2} \frac{(-a)^\ell}{\ell} \{ \mathcal{D}'(\ell) + H_{\ell-1} \mathcal{D}(\ell) \}, \quad (8.3)$$

$$\begin{aligned} \mathcal{J}_3 F_a &= \mathcal{D}'''(0) - 6ad_2 + 3 \sum_{\ell \geq 2} \frac{(-a)^\ell}{\ell} \{ \mathcal{D}''(\ell) + 2H_{\ell-1} \mathcal{D}'(\ell) \\ &\quad + (H_{\ell-1}^2 - H_{\ell-1}^{(2)}) \mathcal{D}(\ell) \}. \end{aligned} \quad (8.4)$$

For two independent Hurwitz parameters, d_1, d_2 are exactly (5.7)–(5.8). For example,

$$\mathcal{D}'(\ell) = \zeta'(\ell, b) \zeta(\ell, c) + \zeta(\ell, b) \zeta'(\ell, c),$$

so every occurrence in (8.3) is explicit. These formulas are exact convergent identities; no assertion is made that the infinite series always collapses to finitely many familiar constants.

8.1 A finite algorithm and its shift continuation

To compute a jet of order r , one first computes the finite Laurent coefficients through d_{r-1} using (5.4), then generates the elementary symmetric coefficients through degree $r-1$, and finally evaluates the normally convergent zeta-jet series. Only finitely many algebraic operations are needed at each series index. The recurrence

$$\ell C_\ell(\mathbf{u}) = \sum_{j=1}^{\ell} (-1)^j \left(\sum_i u_i a_i^j \right) C_{\ell-j}(\mathbf{u}) \quad (8.5)$$

follows by differentiating (2.4) with respect to z and is convenient for multiple factors. The verification code tests it against the independent binomial product (2.5).

The radius $|a| < \lambda_*$ is a convergence condition for this particular Taylor series, not a singularity of the first-jet Gamma representation at positive a . For larger shifts use the finite-head version of the transfer theorem, or use (7.2) and its derivatives. Confusing a convergent expansion's radius with the full identity's domain would unnecessarily discard most positive shifts.

9 Polylogarithmic twists and exact finite-part untwisting

9.1 Twisting lowers the pole order

For $0 < z < 1$ and $c > 0$, the defining Lerch series

$$\Phi(z, s, c) = \sum_{n \geq 0} \frac{z^n}{(n + c)^s}$$

is entire in s , by uniform exponential convergence on compact sets [6]. Consequently the base series of a multiplicative lattice with H untwisted factors and some geometrically twisted factors is

$$\mathcal{D}_z(s) = \prod_{j=1}^H \zeta(s, b_j) \prod_{k=1}^v \Phi(z_k, s, c_k). \quad (9.1)$$

For real $0 < z_k < 1$, its pole order at 1 is exactly H and its leading coefficient is

$$\kappa_{1,H} = \prod_{k=1}^v \Phi(z_k, 1, c_k) > 0.$$

With $H = 0$ it is entire. Complex twists are also covered by the transfer theorem, but a zero of a Lerch factor can lower the pole order further; one must then use the actual principal part, not the number of nominally untwisted factors.

For a particularly transparent arithmetic case, put

$$c_z(N) = \sum_{d|N} z^d d_2(N/d), \quad \mathcal{D}_z(s) = \sum_{N \geq 1} \frac{c_z(N)}{N^s} = \zeta(s)^2 \text{Li}_s(z). \quad (9.2)$$

The equality follows by an absolutely convergent Dirichlet convolution. For $0 < z < 1$, the leading double-pole coefficient is $\text{Li}_1(z) = -\log(1 - z)$. Define $T_{P,z}$ as in (6.4), with $d_2(N)$ replaced by $c_z(N)$. The proof of Theorem 6.1 gives the exact identity

$$\begin{aligned} & T_{P,z}(a, b) + T_{P,z}(b, c) + T_{P,z}(c, a) \\ & - T_{P,z}(b, a) - T_{P,z}(c, b) - T_{P,z}(a, c) = -\frac{\log(1 - z)}{9} (a - b)(b - c)(c - a) \end{aligned} \quad (9.3)$$

for monic quadratic P . This directly connects arithmetic Laurent jets to a classical polylogarithm, while all lower residue coefficients cancel.

9.2 A concrete failure of naive untwisting

At $z = 1$, $c_z(N)$ becomes $d_3(N)$ and the base function is $\zeta(s)^3$. Its order-three pole has subleading coefficient 3γ . For the same *first* Laurent jets and the same quadratic P , alternation kills the degree-one and degree-two parts. The degree-four part is zero because $\mathcal{R}_1$ has degree at most three. The remaining degree-three part gives

$$\text{Alt } T_{P,1}(a, b, c) = \frac{\gamma}{3} (a - b)(b - c)(c - a). \quad (9.4)$$

Here Alt is the signed six-term expression in (9.3). Equations (9.3) and (9.4) exhibit noncommutation without any numerical approximation: the naive $z \uparrow 1$ limit diverges whenever the three shifts are distinct, whereas the untwisted Laurent coefficient is finite. The next theorem supplies the missing compensation at all orders, not only in this example.

9.3 A finite compensation for every polynomial and every jet

Let

$$B(s) = \prod_{j=1}^H \zeta(s, b_j), \quad H \geq 1, \quad Y_n = \prod_{j=1}^H (n_j + b_j), \quad \lambda_{n,n} = nY_n, \quad n \geq 1.$$

The untwisted spectrum has base $\mathcal{D}_0(s) = \zeta(s)B(s)$ and minimum $\lambda_* = \prod_j b_j$. For $t > 0$, attach the weight e^{-tn} to its first factor. The twisted base is

$$\mathcal{D}_t(s) = \text{Li}_s(e^{-t})B(s).$$

For a polynomial $P(x) = \sum_{k=0}^d p_k x^k$, allowed real shifts $a_j > -\lambda_*$, and normalized exponents $\sum \alpha_j = 1$, define

$$Z_{t,P,\alpha}(s) = \sum_{n \geq 1, n \geq 0} e^{-tn} P(\lambda_{n,n}) \prod_j (\lambda_{n,n} + a_j)^{-s\alpha_j} \quad (9.5)$$

initially in its absolute-convergence half-plane and subsequently by continuation. At $t = 0$ omit the exponential weight. Use C_ℓ of (2.4) for these shifts.

Theorem 9.1 (All-order compensated Abel untwisting). *Define the finite meromorphic compensation*

$$M_{t,P,\alpha}(s) = \sum_{k=0}^d p_k \sum_{\ell=0}^{k+1} C_\ell(s\alpha) \Gamma(1-s-\ell+k) t^{s+\ell-k-1} B(s+\ell-k). \quad (9.6)$$

There is a sufficiently small fixed circle $|s| = \rho > 0$ on which

$$Z_{t,P,\alpha}(s) - M_{t,P,\alpha}(s) \longrightarrow Z_{0,P,\alpha}(s) \quad (t \downarrow 0) \quad (9.7)$$

uniformly. In particular, for every integer $r \geq 0$,

$$\boxed{\mathcal{J}_r Z_{0,P,\alpha} = \lim_{t \downarrow 0} \{ \mathcal{J}_r Z_{t,P,\alpha} - r! [s^r] M_{t,P,\alpha}(s) \}. \quad (9.8)}$$

The number of compensation terms depends on the degree of P , not on the requested jet order. For each fixed order its Laurent coefficients are finite expressions in Gamma jets, powers of $\log t$, zeta jets, and generalized Stieltjes constants.

Proof. First suppose $\max |a_j| < \lambda_*$. The binomial expansion gives

$$Z_{t,P,\alpha}(s) = \sum_{k=0}^d p_k \sum_{\ell \geq 0} C_\ell(s\alpha) \text{Li}_{s+\ell-k}(e^{-t}) B(s+\ell-k).$$

Choose $0 < \rho < 1/4$ small enough to avoid any other local poles; the circle never passes through an integer argument of the Gamma factors. The classical polylogarithm expansion is [5, Eq. 25.12.12]

$$\text{Li}_v(e^{-t}) = \Gamma(1-v)t^{v-1} + \sum_{j \geq 0} \zeta(v-j) \frac{(-t)^j}{j!}, \quad 0 < t < 2\pi, \quad (9.9)$$

with colliding terms combined by continuation when v is an integer. For each of the finitely many indices $\ell \leq k+1$, the difference between the left side and its Gamma term tends uniformly on our circle to $\zeta(s+\ell-k)$. Multiplying by the fixed meromorphic factors, which are bounded on the circle, gives exactly the finite subtraction (9.6).

For $\ell \geq k + 2$, no subtraction is needed. The real part of $s + \ell - k$ exceeds 1, and the absolutely convergent series gives

$$\mathrm{Li}_{s+\ell-k}(e^{-t})B(s+\ell-k) \longrightarrow \zeta(s+\ell-k)B(s+\ell-k)$$

uniformly on the circle, by dominated convergence. The high- ℓ bounds from Lemma 2.1 apply uniformly in t , because $0 < e^{-tn} \leq 1$. Together with the Cauchy bound for C_ℓ , they give a common summable geometric majorant. Therefore the entire tail tends uniformly to its untwisted counterpart. This proves (9.7) for small shifts.

For general allowed shifts, remove a finite spectral head so that its tail minimum exceeds $\max |a_j|$. Each subtracted head term is entire in s and converges uniformly to the corresponding untwisted head term as $t \downarrow 0$. Apply the same argument to the tail. The finite subtractions in the base functions converge without a compensation and do not alter the Gamma terms in (9.6). This proves the general case.

Finally, the Laurent coefficient is the Cauchy integral

$$[s^r]F(s) = \frac{1}{2\pi i} \int_{|s|=\rho} \frac{F(s)}{s^{r+1}} ds.$$

Uniform convergence on this fixed circle proves (9.8). Expanding the finitely many Gamma and B factors at their integer arguments gives the stated coefficient class. At a pole of B , its finite required Laurent coefficients are precisely finite Stieltjes polynomials. $\square$

The theorem does not interchange a divergent pointwise limit at $s = 0$ with differentiation. It takes a uniform limit on a contour where the functions are defined, after an explicit finite subtraction. This specifies both the regulator and the correction.

9.4 A closed Bell-polynomial compensation for one shifted factor

For $P = 1$, one shifted factor, and exponent s , the two terms in (9.6) simplify to

$$M_{t,a}(s) = \Gamma(1-s)t^{s-1}B(s) + a\Gamma(1-s)t^sB(1+s). \quad (9.10)$$

The sign of the second term follows from $(-as)\Gamma(-s) = a\Gamma(1-s)$. Write

$$B(1+s) = \sum_{q=-H}^{\infty} b_q s^q, \quad g_j(t) = \left. \frac{d^j}{ds^j} \{\Gamma(1-s)t^s\} \right|_{s=0}.$$

The Gamma expansion [4] gives

$$\Gamma(1-s)t^s = \exp \left((\gamma + \log t)s + \sum_{j \geq 2} \frac{\zeta(j)}{j} s^j \right). \quad (9.11)$$

Thus g_j is the complete exponential Bell polynomial in $\gamma + \log t, 1!\zeta(2), 2!\zeta(3), \dots$; this generating identity also defines it without Bell-polynomial notation.

Corollary 9.2 (Explicit compensation at arbitrary order). *For $r \geq 0$,*

$$\begin{aligned} r![s^r]M_{t,a}(s) &= \frac{1}{t} \sum_{j=0}^r \binom{r}{j} g_j(t) B^{(r-j)}(0) \\ &\quad + ar! \sum_{q=-H}^r b_q \frac{g_{r-q}(t)}{(r-q)!}. \end{aligned} \quad (9.12)$$

Proof. Multiply the Taylor series of (9.11) by the ordinary Taylor series of $B(s)$ in the first term of (9.10), and by its Laurent series $B(1+s)$ in the second. The latter product requires Gamma coefficients through order $r+H$. $\square$

For $B(s) = \zeta(s, c)$, $H = 1$, and $r = 1$, put $\ell_t = \gamma + \log t$. The complete compensation is

$$\frac{\zeta'(0, c) + \ell_t \zeta(0, c)}{t} + a \left\{ \frac{\ell_t^2 + \zeta(2)}{2} + \gamma_0(c) \ell_t - \gamma_1(c) \right\}. \quad (9.13)$$

Subtracting this from $\mathcal{J}_1 Z_{t,1}$ and taking $t \downarrow 0$ yields exactly the Gamma–Stieltjes value (7.2) with parameters $(1, c)$. The $a\ell_t^2/2$ term is indispensable; it comes from the pole of $\zeta(1+s, c)$, and a first-order truncation of the Gamma factor would discard it.

9.5 An explicit compensation check on the six-term family

Return to (9.2), set $z = e^{-t}$, and write $V = (a-b)(b-c)(c-a)$. Alternating the first Laurent coefficient of the finite compensation (9.6), with $B = \zeta^2$ and P monic quadratic, gives

$$\text{Alt}[s]M = -\frac{\log t + 3\gamma}{9} V. \quad (9.14)$$

To check the sign and constant directly, use unnormalized weights $(2, 1, 0)$. The alternated coefficient $C_3(s\mathbf{w})$ is s^3V , while all C_ℓ with $\ell < 3$ have zero alternation. The only relevant compensation is

$$s^3V \Gamma(-3s) t^{3s} \zeta(1+3s)^2.$$

Its first coefficient is (9.14). Hence the compensated signed first jets satisfy the exact finite- t identity

$$\text{Alt } T_{P, e^{-t}} + \frac{\log t + 3\gamma}{9} V = \frac{\gamma}{3} V + \frac{V}{9} \log \frac{t}{1 - e^{-t}}. \quad (9.15)$$

The last logarithm tends to zero. This elementary special case verifies both the divergent subtraction and the finite Euler-constant correction in the general contour argument.

10 Source disposition, verification, and further research

10.1 What has been completed

The question Q10 in [2] is resolved for the concrete classes requested there: periodic and quasi-polynomial multiplicities are handled by (4.1) and Theorem 4.1; several poles and higher poles are handled by Theorems 2.2–3.3 under explicit continuation hypotheses, with unconditional arithmetic examples (4.3) and (5.2). The Hurwitz-product class also supplies all principal coefficients and every individual jet.

This is not a theorem that every nonpolynomial arithmetic multiplicity satisfies those hypotheses. Dirichlet series outside the meromorphic setting, or with a natural boundary blocking the desired spectral point, are not covered by the transfer theorem. Their treatment needs additional structure rather than a formal substitution into its statement.

The previous report already proves the corresponding polynomial and polynomial-times-logarithm calculus. It also already credits the classical multiplicative-anomaly mechanism. Those results are antecedents, not new claims in this package.

10.2 Corrections and integration safeguards

No fresh substantive error is asserted in the targeted current source proofs. The following are proved obstructions to *naïve extensions* and must accompany integration of the new results.

Ordinary derivatives versus Laurent coefficients. Equation (5.14) proves that the double-divisor shift has residue $-a$ at zero. A formula written as $F'_a(0)$ is therefore invalid for $a \neq 0$; the correct object here is $[s]F'_a(s)$. The coordinate is fixed and must be recorded in any identity store.

The full principal part is necessary. The top difference sees only the highest pole, whereas intermediate local identities involve every $\kappa_{p,h}$. At order r one may need the numerator through order $r+H$, not merely $r+1$ or r . The exact multiple-pole tests include a case with a third-order pole at one point and a second-order pole at another.

An Abel limit does not automatically preserve a finite part. The divergence in (9.3) and the finite value in (9.4) prove this explicitly. The compensation in Theorem 9.1 is part of the theorem, not a numerical acceleration that can be dropped.

Do not discard grouped subtractions or base points. The Gamma sums and their primitives converge only in their specified grouped form. The anchored integral vanishes at $a = 0$; an arbitrary negative-polygamma convention cannot replace K without adjusting its polynomial terms.

Preserve the actual Gaussian status. At the inspected revision, the old frozen S_8 candidate is rejected, but a distinct later vector in `04-S8-candidate.tex` remains conjectural. Its certified proximity, like the current S_6 proximity, is not a proof. The present arithmetic-spectrum identities neither prove nor disprove either surviving conjecture. The already proved S_4 identity is not presented as an open target.

10.3 Reproducible algebra and independent numerical diagnostics

The accompanying exact program performs **191** rational or symbolic polynomial equality checks. They include every coefficient of the Pochhammer/harmonic identity for $1 \leq \ell \leq 18$, the full binomial recurrence through degree eight, the alternant normalization through four factors, multiple-pole finite differences, independent Stieltjes product coefficients, and the six-term normalization with a symbolic subleading residue. These finite checks do not substitute for the normal-convergence proofs.

The numerical program uses **60 decimal working digits** and passes **19** equality comparisons. It compares the centered Gamma head plus an analytically summed Hurwitz tail with an independent zeta-product series, repeats that comparison for the primitive, and checks reciprocal polygamma derivatives through order four. It also extracts Laurent coefficients by a discrete Cauchy integral of the unexpanded meromorphic binomial series and compares them with the explicit harmonic/Stieltjes formula through order three. Separate Cauchy extractions check the six-term double-pole, triple-pole, and polylogarithm-twisted identities.

The largest observed absolute discrepancy in this run was less than 6.0×10^{-52} . The fixed pass threshold was 10^{-35} ; the smaller observed residual is a diagnostic, not a rigorously enclosed error. The discrete contour and infinite series were numerically truncated. The mathematical

proofs do not rely on the observed digits. Complete parameters and individual residuals are in `results/numerical_checks.json`.

A separate limiting diagnostic checks (9.13) with $c = 0.9$ and $a = 0.12$. It is deliberately not included among the equality comparisons, because a finite positive t does not give the limiting value.

t	Absolute difference after compensation from the limiting value
10^{-1}	$3.3560095783406 \times 10^{-2}$
10^{-2}	$3.6881187552929 \times 10^{-3}$
10^{-3}	$4.0080828824899 \times 10^{-4}$
10^{-4}	$4.3269361070920 \times 10^{-5}$

The proof of the limit is the uniform-contour argument, not the decreasing sequence of numerical differences.

10.4 Proposed next research questions

The following targets are not promoted to theorems in this report. They are separated from the extension already completed above.

- Q1. Residue-zero strata for complex twists.** For products of Lerch factors, classify the twist parameters where the leading coefficient at $s = 1$ vanishes. Determine the resulting exact degree drops of normalized jets and the transition formulas when several Lerch factors vanish simultaneously. The first test is one simple zero of $\Phi(z, 1, c)$ with a remaining double Hurwitz pole.
- Q2. Several simultaneous untwistings.** Theorem 9.1 removes one geometric twist while retaining an arbitrary Hurwitz product. Construct the finite compensation when two or more twists approach unity at independent rates. Prove which rate-dependent finite terms remain after all divergent powers and logarithms are subtracted. Compare sequential limits with a specified joint limit rather than silently identifying them.
- Q3. Noninteger resonances at other spectral centers.** At the origin, only integer poles $p = \ell - k$ enter the local formula. Develop the corresponding finite local calculus about a nonzero total spectral center so that noninteger or complex poles can resonate. State the exponent normalization and branch choices before defining its tangent polynomial law.
- Q4. A smaller coordinate algebra for centered Gamma reciprocity.** For specified rational a, b, c , determine which values of $E_{b,c}(a)$ and $I_{b,c}(a)$ reduce to a finite algebra generated by Gamma values, Hurwitz-zeta derivatives, classical or multiple polylogarithms, and Dirichlet L derivatives. The comparison algebra must be fixed. Failure of a symbolic simplifier or a relation search would not prove irreducibility.
- Q5. A reflection or functional equation for the shifted divisor spectrum.** Construct a useful exact reflection formula in the shift variable for $F_a(s)$ or its Laurent jets, rather than for the unshifted product $\zeta(s)^h$. The Gamma representation supplies ordinary derivatives, but does not by itself give a global reflection law across the negative spectral lattice.
- Q6. Canonical finite-part determinants with several pole orders.** Our choice $\exp(-[s]F(s))$ is well-defined once the spectral coordinate is fixed. Determine a normalization with a specified covariance under spectral rescaling and a specified product

law. Record the lower Laurent coefficients that necessarily enter under a coordinate change; there is no regulator-independent determinant for free.

- Q7. Representation components beyond the Vandermonde.** For a pole of order H , the normalized jet has degree at most $r + H$. Identify permutation-representation projections whose first occurrence lies above degree r but below the fully alternating degree. They should isolate particular Stieltjes residue layers. Give explicit factorizations and exact finite certificates in the first four-factor cases.
- Q8. Other arithmetic products with controlled continuation.** Apply the transfer theorem to products of primitive Dirichlet L functions and shifted zeta factors, tracking zeros that cancel nominal poles. Separate residue-class identities from new arithmetic information. For a base series lacking known continuation, prove that continuation rather than treating the theorem as evidence for it.
- Q9. Formal analytic interfaces and certified regression tests.** Formalize the finite-principal-part extraction and tangent polarization independently of the convergence lemmas. Add interval Cauchy extraction and certified truncation to the current floating-point diagnostics. The exact finite algebra is already isolated in the supplied program.
- Q10. The surviving Gaussian identities remain a separate target.** An exact certificate for S_6 , and then the distinct current S_8 vector, still needs a relation outside the restricted product-row presentation identified in the source. The present divisor-spectrum alternants should not be mistaken for such a certificate.

10.5 Integration recommendation

Place the package under a new continuation directory beneath the Stieltjes-harmonic-resonance report, and mark the explicit Q10 classes as completed with the hypotheses and scope above. The main candidate manuscript insertions are the transfer/local calculus, the Gamma reciprocity with its normalized primitive, and the compensated polylogarithmic untwisting theorem. The package is self-contained: its $\text{\LaTeX}$ build does not depend on unbundled repository files. No remote repository files were modified while preparing this report.

References

- [1] ProveIt repository, *Polylogarithms research programme*, canonical manuscript and incoming-directory inventory, targeted snapshot 3fdd6cc447fac0731801588723109ec3b2e21cec. Relevant chapters: 07-integration.tex, 08-differentiation.tex, 10-discovery.tex, and 04-S8-candidate.tex. <https://github.com/VladimirReshetnikov/ProveIt/tree/3fdd6cc447fac0731801588723109ec3b2e21cec/Analysis/Polylogarithms/docs/manuscript>.
- [2] ProveIt research continuation, *Resonant Gamma Jets and Directional Zeta Identities: Discrete Stieltjes products, shifted polylogarithms, and local formulas for higher spectral derivatives*, 10 October 2026. Especially Section 7, “Local identities for every derivative of a spectral product,” and Section 8, question Q10. <https://github.com/VladimirReshetnikov/ProveIt/tree/3fdd6cc447fac0731801588723109ec3b2e21cec/Analysis/Polylogarithms/docs/reports/stieltjes-harmonic-resonance/continuations/resonant-jets-and-shifts>.
- [3] NIST Digital Library of Mathematical Functions, *Section 25.11: Hurwitz Zeta Function*. Definitions, continuation, parameter derivatives, and relations to Gamma and polygamma; accessed during preparation. <https://dlmf.nist.gov/25.11>.
- [4] NIST Digital Library of Mathematical Functions, *Section 5.7: Series Expansions of the Gamma Function*. <https://dlmf.nist.gov/5.7>.
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- [7] J. S. Dowker, *Calculation of the multiplicative anomaly*, arXiv:1412.0549, 2014; corrected version 3, 24 September 2023. <https://arxiv.org/abs/1412.0549>.